Karishma Premnath

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Education

Northeastern University, Boston, MA — Jan 2026 – Present
Master of Science in Electrical and Computer Engineering — Expected May 2028
Relevant Coursework: Machine Learning, Robotics — **GPA: 3.834/4.00**

Sastra Deemed University, Thanjavur, Tamil Nadu, India — Oct 2021 – Sept 2025
Bachelor of Technology in Electronics and Instrumentation Engineering

Skills

Control & Automation: PLC programming (IEC 61131-3 Ladder Diagram, Structured Text), PID and loop tuning, cascade and ratio control, Model Predictive Control, SCADA and HMI, Modbus TCP
Instrumentation & Standards: Transducers, instrument index and I/O list, Sensor selection and ranging, 4-20 mA loop design, signal conditioning and calibration, data acquisition, system identification, ISA-5.1 (P&ID), ISA-18.2 (alarm management), troubleshooting and commissioning, Factory Acceptance Test (FAT)
Programming & Tools: C, Structured Text, Embedded C, SQL, Python, CODESYS, OpenPLC, MATLAB, Simulink, AutoCAD, LabVIEW, Arduino IDE, Multisim

Work Experience

Apollo Hospital | Biomedical Engineering Intern | Trichy, Tamil Nadu, India — June 2025 – July 2025
- Maintained and serviced 25+ diagnostic instruments deployed across patient care units, where one failed sensor takes a device out of service
- Assisted a senior engineer in diagnosing and resolving 6 medical instruments through hardware-software interfacing, sensor connectivity and signal validation
- Identified recurring hardware connectivity issues and inconsistent signal thresholds for 3 medical devices, restoring all 3 to operation

Dalmia Cements | Process Control & Automation Intern | Dalmiapuram, Tamil Nadu, India — July 2024
- At a cement production unit running crushing, pyroprocessing and cement packing, monitored 10 process parameters through PLC and SCADA/HMI dashboards
- Traced 10 field instruments across 3 production stages, recording signal type and measurement range for each
- Documented 15 process variations along with sensor connectivity data, with no wiring faults found

CodeBind Technologies | Embedded Systems Intern | Chennai, Tamil Nadu, India — June 2023
- Manual tracking of 6 parking slots took 3 minutes per scan and caused delays in vehicle entry. Designed and deployed an automated car parking system using Arduino, IR sensors, and embedded C, incorporating sensor hardware with control software and an Android app
- Minimized the tracking time to 10 seconds, an 18x reduction, with correct availability logged on 20 of 23 test runs

Projects

Water Treatment Skid: PLC Control System Design and Testing — Aug 2026
- Built and tested a PLC control system for two-stage water treatment by specifying 4 PID loops (cascade level-flow and ratio control), 12 alarms, 8 interlocks and 33 I/O signals (7 AI, 4 AO, 14 DI, 8 DO)
- Developed IEC 61131-3 ladder logic and structured text (ST), Modbus TCP plant simulation and process HMI. Executed the 14 FAT cases and 26 logic tests, found 6 design defects across 40 test cases, rectified all 6 for 100% pass

ML-Assisted PLC Structured Text Bug Detection and Rule-Based Repair — June 2026
- Developed a dataset of 300 PLC structured text (ST) programs featuring 8 functional defects, such as disabled interlock and incorrect setpoint, that compile successfully but lead to failures during operation
- Implemented a Python workflow for automated feature extraction from code, then trained a decision tree model to predict the functional errors. Elevated defect classification accuracy from 81.7% baseline to 100% over 60 held-out programs and rectified all 60 single-defect test programs

Smart Wearable Respiratory & Heart Rate Monitoring System — May 2024
- Built wearable belt for monitoring heart rate and respiratory rate from single force-sensitive resistor (FSR) as conventional monitoring needs separate transducers
- Implemented and calibrated the analog signal chain with I-V amplification across a 16-point load curve from 0.98 to 19 N, an LM324 active low pass filter (0.45 Hz) for respiration and a band pass filter (2.5-5 Hz) for the ballistocardiogram, evaluated within 48-108 BPM and 8-23 breaths/min

Publication

Model predictive controller design using machine learning models & implementation of parameter tuning
IEEE AIDE 2026: Technically Co-Sponsored Conference, NMAM Institute of Technology, Nitte, India — Jun 2026
- Implemented and validated a model predictive controller on a laboratory temperature control station, evaluating 4 machine learning models (KNN, feedforward neural network (FNN), ridge regression and decision tree) against 4 conventional methods (state-space, transfer function, ARX and Box-Jenkins) across varying tuning parameters
- Ridge-based MPC achieved the lowest MSE of 0.0058 and ITAE of 12.2788 among the evaluated models. Published in IEEE Xplore. ieeexplore.ieee.org/document/11545113